

Logic

1. Determine if the statement is true or false. If false, give a counterexample.

$$\forall a, b \in \mathbb{R}, a|b \rightarrow a^2|b^2$$

2. Prove or Disprove: If $p \rightarrow q$ and $q \rightarrow r$, then $(p \vee q) \rightarrow r$

Cartesian Products

1. If $|A| = n$ and $|B| = m$, prove $|A \times B| = nm$
2. How many subsets of $|A \times B|$ exist? Assume $|A| = n$, $|B| = m$, and $|A \cap B| = 0$.

Set Theory

1. Prove that $(A \setminus B) \cap B = \emptyset$
2. Assume $|A| = 10$, $|B| = 30$, and $|A \cap B| = 10$. What can we infer about A and B ?

Functions

1. If $G : A \rightarrow B$ and $F : B \rightarrow C$, prove $F \circ G : A \rightarrow C$ is injective.
2. If $F : A \rightarrow B$ is bijective, prove that F^{-1} exists and is bijective.

Number Theory

1. Prove that 101 is prime.
2. Prove that if $\gcd(a, b) = 1$, then $a|bc \rightarrow a|c$

Division Theorem

1. Write -57 in the form: $8q + r$ with $0 \leq r < 8$
2. Prove uniqueness of the division theorem. (Existence was previously proven in class)

Euclid's Algorithm

1. Find integers x, y s.t. $414x + 662y = \gcd(414, 662)$
2. Find multiplicative inverse of $37 \pmod{101}$

Modular Arithmetic and Fermat's Little Theorem

1. Determine if 18 has an inverse modulo 30
2. Compute $7^{100} \pmod{13}$
3. Prove that if p is prime and $p \nmid a$, then $a^{p-2} = a^{-1} \pmod{p}$

Equivalence Classes and CRT

1. How many equivalence classes are there under mod n on $\mathbb{Z}$?
2. Find integer x s.t. $x \equiv 1 \pmod{4}$ and $x \equiv 3 \pmod{6}$. Solve or explain why it is impossible.

Induction

1. We define the fibonacci sequence s.t. $n_1 = 0, n_2 = 1$ and $\forall n > 2, a_n = a_{n-1} + a_{n-2}$. Prove that $a_n < 2^n \forall n \geq 1$.

Wilson's Theorem

1. Prove both direction of Wilson's Theorem. In other words, prove that p is prime $\iff (p-1)! \equiv -1 \pmod{p}$